MA1522 Linear Algebra for Computing

Lesson 9: Subspaces and Linear Independence

Announcement: Online Quiz 2

- **Quiz 2 covers Chapter 2**, Sections 2.1–2.10: matrix algebra, matrix equations, inverses and elementary matrices, LU factorization, and determinants.

- It opens **Saturday 12 September, 0:00** and is due **Sunday 13 September, 23:59**.

30 mins. 10 multiple choice questions.

Submission

- You only need to submit your attempts at those labelled **Question 1** to **Question 3**.
- If you have written your answer on A4 paper, scan or take pictures of your work (make sure the images can be read clearly). Merge all your images into one PDF file, in page order.
- If you have written or typed your answer digitally, print it into a PDF file.
- Name the PDF file by `StudentNo_lesson9` (e.g. `A1234567Z_lesson9.pdf`).
- Submit the PDF file to Canvas assignments, **Participation(L1): Lesson 9**.
- Deadline: today, **12:30 PM**.
- If you do not know how to upload your submission, come to ask me after class.

Pollev



Example

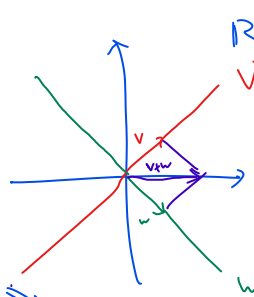
Span(S) = V Span(T) = W

Let V and W be subspaces of $\mathbb{R}^n$. True or false?

1. $V \cup W$ is a subspace of $\mathbb{R}^n$.
2. $V \cap W$ is a subspace of $\mathbb{R}^n$. $= \text{Span}(\,?\,)$.

Consider $V = \text{span}\left\{\begin{pmatrix}1\\1\end{pmatrix}\right\}$, $W = \text{span}\left\{\begin{pmatrix}1\\-1\end{pmatrix}\right\}$.

$V + W = \begin{pmatrix}2\\0\end{pmatrix} \notin V \cup W$



$\Rightarrow$ $V \cup W$ is not a subspace.

(i)': $V \cap W \neq \emptyset$.

because $\begin{cases} V \text{ contains } 0 \\ W \text{ contains } 0 \end{cases} \Rightarrow V \cap W \ni 0$

(ii)': for $u_1, u_2 \in V \cap W$, $\alpha, \beta \in \mathbb{R}$, we check $\alpha u_1 + \beta u_2 \in V \cap W$.

- $\alpha u_1 + \beta u_2 \in V$ since $u_1, u_2 \in V$
- $\alpha u_1 + \beta u_2 \in W$ since $u_1, u_2 \in W$.

$\Rightarrow \alpha u_1 + \beta u_2 \in V \cap W$

Section 3.4 Computing $V \cap W$

Let $V = \text{span}\{\mathbf{u}_1, \ldots, \mathbf{u}_k\}$ and $W = \text{span}\{\mathbf{w}_1, \ldots, \mathbf{w}_l\}$ be subspaces of $\mathbb{R}^n$. Then

$$\begin{aligned}
\mathbf{v} \in V \cap W &\iff \mathbf{v} \in V \text{ and } \mathbf{v} \in W \\
&\iff \mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k = d_1\mathbf{w}_1 + \cdots + d_l\mathbf{w}_l \text{ for some } c_i, d_j \in \mathbb{R} \\
&\iff \begin{pmatrix} \mathbf{u}_1 & \cdots & \mathbf{u}_k & \mathbf{w}_1 & \cdots & \mathbf{w}_l \end{pmatrix} \begin{pmatrix} c_1, & \ldots, & c_k, & -d_1, & \ldots, & -d_l \end{pmatrix}^T = \mathbf{0}.
\end{aligned}$$

So: solve that homogeneous system, then read the first k entries of each solution and substitute back,

$$\mathbf{v} = c_1\mathbf{u}_1 + \cdots + c_k\mathbf{u}_k.$$

Question 1

Let $V = \text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$ and $W = \text{span}\{\mathbf{w}_1, \mathbf{w}_2\}$ be subspaces of $\mathbb{R}^3$, where

$$\mathbf{u}_1 = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}, \quad \mathbf{w}_1 = \begin{pmatrix} 2 \\ 2 \\ 3 \end{pmatrix}, \quad \mathbf{w}_2 = \begin{pmatrix} 2 \\ 1 \\ 1 \end{pmatrix}.$$

Compute $V \cap W$. = Span (. . .)

Question 1 — solution

$v \in \mathrm{Span}(\{u_1, u_2\}) \cap \mathrm{Span}(\{w_1, w_2\})$ $\Big|$ (*) $\iff$ $v = c_1u_1 + c_2u_2 = d_1w_1 + d_2w_2$ for some $c_1, c_2, d_1, d_2 \in \mathbb{R}$.

$\iff$ $c_1u_1 + c_2u_2 - d_1w_1 - d_2w_2 = 0$.

$\iff$ $\begin{pmatrix} | & | & | & | \\ u_1 & u_2 & w_1 & w_2 \\ | & | & | & | \end{pmatrix} \begin{pmatrix} c_1 \\ c_2 \\ -d_1 \\ -d_2 \end{pmatrix} = 0$.

a linear system.

$\left(\begin{array}{cccc|c} 1 & 3 & 2 & 2 & 0 \\ 1 & 2 & 2 & 1 & 0 \\ 2 & 2 & 3 & 1 & 0 \end{array}\right)$ rref $\left(\begin{array}{cccc|c} 1 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & -1 & 0 \end{array}\right)$

general sol. $\begin{pmatrix} c_1 \\ c_2 \\ -d_1 \\ -d_2 \end{pmatrix} = \begin{pmatrix} -t \\ -t \\ t \\ t \end{pmatrix}$

set $-d_2 = t$, $-t(u_1+u_2) = -t(w_1+w_2)$

(*) reads: $v \in V \cap W \iff v = -tu_1 - tu_2 = -tw_1 - tw_2$ for some $t \in \mathbb{R}$.

$\iff$ $v \in \mathrm{Span}(\{u_1 + u_2\})$

$= \{ s \cdot (u_1+u_2) \mid s \in \mathbb{R} \}$.

Ans: $V \cap W = \mathrm{span}\left(\left\{\begin{pmatrix} 4 \\ 3 \\ 4 \end{pmatrix}\right\}\right)$

Challenge

Let V and W be subspaces of $\mathbb{R}^n$. Show that

$$V \cup W \text{ is a subspace of } \mathbb{R}^n \iff V \subseteq W \text{ or } W \subseteq V.$$

Challenge: $V_1, V_2, V_3 \leq \mathbb{R}^n$ be subspaces.

$V_1 \cup V_2 \cup V_3$ is a subspace $\iff \exists i$ s.t. $V_i \supseteq V_1$
$V_i \supseteq V_2$
$V_i \supseteq V_3$.

Section 3.5 Linear Independence

Definition

A set $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ is **linearly independent** if the only coefficients $c_1, c_2, \ldots, c_k$ satisfying

$$c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_k\mathbf{u}_k = \mathbf{0} \implies c_1 = 0, c_2 = \cdots = c_k$$

are $c_1 = c_2 = \cdots = c_k = 0$. Otherwise the set is **linearly dependent**.

A set is linearly dependent exactly when one of its vectors is a linear combination of the others: if $c_i \neq 0$ in such a relation, then

$$\mathbf{u}_i = \frac{c_1}{-c_i}\mathbf{u}_1 + \cdots + \frac{c_{i-1}}{-c_i}\mathbf{u}_{i-1} + \frac{c_{i+1}}{-c_i}\mathbf{u}_{i+1} + \cdots + \frac{c_k}{-c_i}\mathbf{u}_k.$$

$\frac{c_1}{-c_i}\mathbf{u}_1 + \cdots + \frac{c_{i-1}}{-c_i}\mathbf{u}_{i-1} + \frac{c_{i+1}}{-c_i}\mathbf{u}_{i+1} + \cdots + \frac{c_k}{-c_i}\mathbf{u}_k = \frac{-c_i}{-c_i}\mathbf{u}_i$.

$\therefore \mathbf{u}_i \in \mathrm{Span}(S - \{\mathbf{u}_i\})$

Exercise: If $v \in \mathrm{Span}(S)$, then $\mathrm{Span}(S \cup \{v\}) = \mathrm{Span}(S)$.

Section 3.5 Checking Linear Independence

$(u_1 \cdots u_k)\begin{pmatrix} c_1 \\ \vdots \\ c_k \end{pmatrix} = 0$

Theorem

Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$ be a set of vectors in $\mathbb{R}^n$ and let $A = \begin{pmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_k \end{pmatrix}$. The following are equivalent:

(i) S is linearly independent;

(ii) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution;

(iii) the reduced row-echelon form of A has no non-pivot column.

Three special cases:

1. $\{\mathbf{0}\}$ is linearly dependent.
2. $\{\mathbf{v}\}$ is linearly independent if and only if $\mathbf{v} \neq \mathbf{0}$.
3. $\{\mathbf{v}_1, \mathbf{v}_2\}$ is linearly dependent if and only if one is a scalar multiple of the other.

Question 2

For each of the following sets, determine whether it is linearly independent. If it is not, exhibit a linearly independent subset with the same linear span.

1. $S_1 = \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 2 \end{pmatrix} \right\}$

$A = \left(\begin{array}{ccc|c} | & | & | & 0 \\ u_1 & u_2 & u_3 & 0 \\ | & | & | & 0 \end{array} \right) \xrightarrow{\text{ref.}} \left(\begin{array}{ccc|c} 1 & & & 0 \\ & 1 & & 0 \\ & & 1 & 0 \end{array} \right) \Rightarrow$ linearly independent.

2. $S_2 = \{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4\}$, where

$$\mathbf{v}_1 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}, \quad \mathbf{v}_2 = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{v}_3 = \begin{pmatrix} 1 \\ 1 \\ -3 \end{pmatrix}, \quad \mathbf{v}_4 = \begin{pmatrix} 1 \\ 4 \\ 3 \end{pmatrix}.$$

Question 2 — solution: S_1

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 3 & 1 & 4 \\ -1 & 1 & -3 & 3 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix} \Longrightarrow$$

u_1 u_2 u_3 u_4

NOT linearly independent since there are non-pivot columns.

Next: which vectors should we remove?
Equivalently, which vectors can be expressed as a linear combination of the others?

The systems

$$\begin{cases} (u_1 \; u_2)\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = (u_3) & c_1 = 2,\ c_2 = -1 \\ (u_1 \; u_2)\begin{pmatrix} c_1 \\ c_2 \end{pmatrix} = (u_4) & c_1 = -1,\ c_2 = 2. \end{cases}$$

are consistent. $\iff u_3, u_4 \in \text{Span}(\{u_1, u_2\})$.

Therefore, $\text{Span}(S_2) = \text{Span}(\{u_1, u_2\})$.

Finally, $\{u_1, u_2\}$ is linearly independent since

$$\begin{pmatrix} 1 & 1 \\ 2 & 3 \\ -1 & 1 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Question 2 — solution: S_2

A general fact: $S = \{u_1, \ldots, u_k\}$.

$$A = \begin{pmatrix} | & & | \\ u_1 & \cdots & u_k \\ | & & | \end{pmatrix} \xrightarrow{\text{ref}} \begin{pmatrix} | & & | \\ w_1 & \cdots & w_k \\ | & & | \end{pmatrix}.$$

Let $\{w_{i_1}, w_{i_2}, \ldots, w_{i_s}\}$ be the pivot columns,

then $\text{Span}(S) = \text{Span}(\{u_{i_1}, u_{i_2}, \ldots, u_{i_s}\})$.

Question 3

Let $\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}$ be a linearly independent set of vectors in $\mathbb{R}^n$. Determine whether each of the following sets is linearly independent.

1. $S = \{\ \mathbf{u} - \mathbf{v},\ \ \mathbf{v} - \mathbf{w},\ \ \mathbf{w} - \mathbf{u}\ \}$

2. $T = \{\ \mathbf{u} + \mathbf{v},\ \ \mathbf{v} + \mathbf{w},\ \ \mathbf{w} + \mathbf{u}\ \}$

Question 3 — solution: S

$\{u-v,\ v-w,\ w-u\}$

Suppose $c_1(u-v) + c_2(v-w) + c_3(w-u) = 0$. What are the conditions on c_i?

$\Updownarrow$

$(c_1 - c_3)\cdot u + (-c_1+c_2)\,v + (-c_2+c_3)\,w = 0$ $\xrightarrow[\text{of } \{u,v,w\}]{\text{linearly indep.}}$ $\begin{cases} c_1 - c_3 = 0 \\ -c_1 + c_2 = 0 \\ -c_2 + c_3 = 0. \end{cases}$

Solve $\begin{pmatrix} 1 & 0 & -1 \\ -1 & 1 & 0 \\ 0 & -1 & 1 \end{pmatrix} \xrightarrow{\text{rref}} \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$.

no unique solution.

In fact, $\begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$ gives a non-trivial sol.

I.e. $1\cdot(u-v) + 1\cdot(v-w) + 1\cdot(w-u) = 0$

$\Rightarrow \{u-v,\ v-w,\ w-u\}$ is not l.I.

Question 3 — solution: T

$\{u+v,\ v+w,\ w+u\}$.

Suppose

$$c_1(u+v) + c_2(v+w) + c_3(w+u) = 0$$

$$\Downarrow$$

$$(c_1+c_3)u + (c_1+c_2)v + (c_2+c_3)w = 0$$

$\Longrightarrow$ l.I. of $\{u, v, w\}$ $\Longrightarrow$

$$\begin{cases} c_1 + c_3 = 0 \\ c_1 + c_2 = 0 \\ c_2 + c_3 = 0 \end{cases}$$

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{array}\right) \xrightarrow{\text{rref}} \left(\begin{array}{ccc|c} 1 & & & 0 \\ & 1 & & 0 \\ & & 1 & 0 \end{array}\right)$$

This means that $c_1 = c_2 = c_3 = 0$ is the unique sol.

$\Longrightarrow \{u+v,\ v+w,\ w+u\}$ is linearly independent. $\square$

Next lesson

- **Sections 3.6 & 3.7 Basis and Coordinates, Dimensions**

Please watch the video clips for these sections beforehand.